Ref Zero

AI-Powered Sports Referee Assistant

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Abstract

Ref Zero is an end-to-end multimodal AI system that acts as a real-time Video Assistant Referee (VAR) for sports. We fine-tune a YOLO11n model on a curated 353-image sports foul detection dataset achieving mAP@0.5 of 0.554, then integrate it with a MediaPipe pose estimation overlay, a Smart DVR buffer, and Gemini 2.0 Flash for natural language rule citation. When a foul is detected with confidence above 0.5, the system saves a 5-second clip and sends it to Gemini, which returns a structured JSON verdict citing specific rulebook rules (e.g., "NBA Rule 12B — Blocking Foul"). A browser-use agent then autonomously verifies the citation against the official rulebook. The full system achieves end-to-end review in under 10 seconds, compared to 2-5 minutes for traditional VAR. Our MLOps pipeline includes W&B experiment tracking, an ablation study across three YOLO11 model sizes, GitHub Actions CI/CD, and deployment to Hugging Face Spaces.

1. Introduction

Every year, controversial referee calls cost teams championships, millions in revenue, and fans' trust. The NFL employs 450+ officials across 256 games, the NBA regularly faces missed calls that swing playoff games, and soccer's VAR system still relies heavily on human interpretation. Youth and amateur sports — representing 90% of all games played — have no video review capability whatsoever.

Current video review systems require referees to stop the game for 2-5 minutes, manually scrub footage frame-by-frame, recall rules from 200+ page rulebooks, and make judgment calls under significant pressure. Even with these tools, human error remains prevalent.

We propose Ref Zero: an AI referee assistant that gives every referee superhuman capabilities. The system continuously processes live video using computer vision, automatically captures critical moments, and uses a fine-tuned YOLO11 model to detect fouls in real time. When a potential foul is detected, Gemini 2.0 Flash analyzes a 5-second clip and returns a structured verdict with specific rule citations, which are then autonomously verified against official rulebooks by a browser-use agent.

Our key contributions are: (1) a fine-tuned YOLO11n sports foul detection model achieving mAP@0.5 = 0.554 on a curated dataset, (2) a complete MLOps training pipeline with W&B experiment tracking and ablation studies, (3) a multi-modal inference pipeline combining YOLO11, MediaPipe, and Gemini 2.0 Flash, and (4) an agentic rulebook verification system using browser-use.

2. Related Work

2.1 Object Detection for Sports Analysis

Real-time object detection in sports has been an active research area. Redmon et al. (2016) introduced YOLO, establishing the single-stage detection paradigm that prioritizes inference speed — critical for real-time applications. Subsequent iterations improved accuracy-speed tradeoffs. Ultralytics YOLO11 (2024) introduces C3k2 blocks and an improved detection head, achieving state-of-the-art results on COCO while maintaining real-time inference speeds suitable for broadcast video analysis.

2.2 Pose Estimation in Sports

Cao et al. (2019) demonstrated real-time multi-person pose estimation using Part Affinity Fields, enabling player body tracking across sports broadcasts. Google's MediaPipe (Lugaresi et al., 2019) provides production-ready pose estimation tracking 33 body landmarks at 30+ FPS, which we use for real-time skeleton overlay in Ref Zero. Pose information complements object detection by providing contextual body position data.

2.3 Video Understanding with Large Language Models

The emergence of multimodal large language models has enabled video understanding capabilities beyond traditional computer vision. Google's Gemini 2.0 Flash supports native video input and can reason about temporal sequences of frames — making it suitable for analyzing sports plays where context across multiple frames determines the correct ruling.

2.4 Agentic AI Systems

Browser-use (2024) enables LLM agents to autonomously navigate web interfaces. We leverage this for autonomous rulebook verification — the agent opens official sports federation websites and extracts the specific rule text cited by Gemini, providing traceable evidence for each verdict. This represents a novel application of agentic AI to sports officiating.

2.5 VAR in Professional Sports

Video Assistant Referee systems have been deployed in professional soccer since 2018. Studies show VAR reduces clear refereeing errors by 79% but increases game interruptions significantly (Lago-Penas et al., 2021). Our system addresses this by automating the analysis pipeline, targeting a reduction from 5-minute reviews to under 10 seconds.

3. Data

3.1 Dataset Source

We use the Foul Detection Dataset from Roboflow Universe (CC BY 4.0 license), created specifically for basketball foul detection. The dataset contains 353 labeled images across three classes: foul, no-foul, and contact.

Split	Images	Percentage	Use

Train	247	70%	Model parameter optimization
Validation	71	20%	Hyperparameter tuning & early stopping
Test	35	10%	Final evaluation (held out)

3.2 Class Distribution

Class	Description	Test Images	Test Instances
foul	Illegal contact resulting in a foul call	23	23
no-foul	Legal play or incidental contact	21	21
contact	Physical contact, may or may not be foul	41	83

3.3 Data Augmentation

Given the small dataset size (353 images), augmentation is critical to prevent overfitting. We apply the following augmentation pipeline, configured in backend/model_config.py:

Augmentation	Value	Justification
Mosaic	1.0 (always)	Combines 4 images per batch — quadruples effective diversity without additional labeling
MixUp	0.1	Adds regularization; low value avoids blurring foul/no-foul boundaries
Horizontal Flip	0.5	Fouls occur on both sides of court — doubles effective dataset
Vertical Flip	0.0 (disabled)	Upside-down players don't occur in real broadcasts
HSV Jitter	h=0.015, s=0.7, v=0.4	Handles court/jersey color variations and lighting conditions
Rotation	±5°	Simulates camera tilt; beyond 10° is unrealistic

4. Methods

4.1 System Architecture

Ref Zero implements a five-stage pipeline:

- Stage 1 — Computer Vision: Live camera feed processed by MediaPipe at 30 FPS, tracking 33 body keypoints per player for real-time skeleton overlay
- Stage 2 — Foul Detection: Fine-tuned YOLO11n model analyzes each frame with confidence threshold 0.5. On foul/contact detection, triggers the DVR buffer
- Stage 3 — Smart DVR: Rolling 150-frame buffer (5 seconds at 30 FPS) saves the last 5 seconds as MP4 when triggered

- Stage 4 — AI Analysis: Gemini 2.0 Flash receives the clip and returns structured JSON with sport, action breakdown, rule violated, verdict, and confidence score
- Stage 5 — Rule Verification: browser-use agent autonomously opens official rulebook URLs and verifies the cited rule text

4.2 YOLO11 Model Architecture

We fine-tune YOLO11n (nano variant) on our foul detection dataset. YOLO11 introduces several improvements over YOLOv8: C3k2 blocks replace C2f in the backbone, providing better feature extraction with fewer parameters; an improved C2PSA attention module in the neck; and enhanced detection head with better small-object localization. These improvements make YOLO11n well-suited for detecting contact zones between players, which represent small regions within broadcast video frames.

The model is initialized with COCO pre-trained weights, then fine-tuned for 50 epochs on our dataset. We use the AdamW optimizer with cosine learning rate decay from lr0=0.001 to lrf=0.01 × lr0, with 3 warmup epochs to prevent early instability. All training was performed on Apple M2 Pro MPS (Metal Performance Shaders) completing in 2.713 hours.

4.3 Loss Function Design

Loss Component	Weight	Formula	Justification
Box (CIoU)	7.5	$\text{CIoU} = \text{IoU} - \rho^2/c^2 - \alpha v$	Penalises position, overlap, AND aspect ratio. Foul boxes are tall/thin — aspect ratio error matters
Classification (BCE)	0.5	BCE with logits	Multi-label safe — a frame can simultaneously be 'contact' AND 'foul'
Distribution Focal	1.5	DFL on boundary distribution	Improves regression on ambiguous contact edges where foul boundaries are uncertain

4.4 Hyperparameter Configuration

Parameter	Value	Justification
Input size	640px	Balances detection accuracy vs. FPS — 1280px gave +2% mAP but dropped below real-time threshold
Batch size	16	Fits M2 Pro MPS memory; larger batches caused OOM
Epochs	50	Sufficient convergence for 353-image dataset; early stopping (patience=10) prevents overfit
Optimizer	AdamW	Outperforms SGD on small datasets via adaptive learning rates + weight decay
LR	0.001 → 0.00001	Standard YOLO fine-tuning range with cosine decay

Confidence threshold	0.5	Tuned via W&B sweep — 0.3 caused false positives, 0.7 missed real fouls
Activation	SiLU	YOLO11 default — smoother gradients than ReLU, differentiable at zero
Normalisation	BatchNorm	Stabilises training on small batches; faster convergence than InstanceNorm

4.5 MLOps Pipeline

Our training pipeline implements MLOps Level 2 with the following components:

- Experiment Tracking: W&B logs all metrics (box loss, cls loss, dfl loss, mAP@0.5, mAP@0.5-0.95, precision, recall) per epoch
- Model Registry: Best checkpoint (best.pt) selected by validation mAP@0.5 with path written to runs/best_model.txt for automatic hot-swapping
- MLflow Integration: Ultralytics auto-logs runs to runs/mlflow for additional experiment management
- CI/CD: GitHub Actions workflow triggers 5-epoch training on push to main, uploads checkpoint as artifact
- Containerization: Docker + docker-compose for one-command reproducible deployment
- Hot-swap: POST /api/retrain endpoint retrains in background and automatically reloads new checkpoint without server restart

5. Experiments

5.1 Main Results — YOLO11n (50 epochs)

Class	Precision	Recall	mAP@0.5	mAP@0.5-0.95
all (overall)	0.463	0.651	0.554	0.281
foul	0.397	0.609	0.527	0.231
no-foul	0.565	0.742	0.606	0.393
contact	0.426	0.602	0.527	0.219

The model achieves 0.554 mAP@0.5 overall. The no-foul class achieves the highest performance (mAP50=0.606) as it represents the largest proportion of negative examples and the model correctly identifies clean play. The foul and contact classes show similar performance, which is expected as many contact instances are on the boundary between categories. Inference runs at 226.9ms/image on CPU and significantly faster on MPS.

5.2 Ablation Study — Model Size

Model	Params	mAP@0.5	Precision	Recall	Train Time	Checkpoint Size	Decision
YOLO11n ✓	2.6M	0.554	0.463	0.651	2.71h (50ep)	5.5MB	SELECTED
YOLO11s	9.4M	0.491	0.590	0.495	0.66h (10ep)	19.2MB	Rejected
YOLO11m	20M	N/A	—	—	Timeout	40.5MB	Too large

YOLO11n achieves the highest mAP@0.5 (0.554) despite being the smallest model, demonstrating that the COCO pre-training provides strong transfer learning even with nano architecture. YOLO11s, trained for only 10 epochs (vs. 50 for nano), underperforms — indicating it would likely reach higher accuracy with full training. YOLO11m timed out on Apple M2 Pro hardware before completing even one epoch, confirming that medium-scale models are unsuitable for edge deployment and real-time refereeing scenarios requiring >30 FPS.

The ablation validates our production choice: YOLO11n provides the best accuracy-speed-size tradeoff for real-time refereeing requirements.

5.3 Hyperparameter Sweep

We ran a W&B grid sweep over 12 combinations of learning rate (0.001, 0.0005, 0.0001), batch size (8, 16), and confidence threshold (0.3, 0.5), optimizing for validation mAP@0.5. The optimal configuration was lr0=0.001, batch=16, conf=0.5 — confirming our default hyperparameter choices in model_config.py.

Key findings: (1) lr=0.001 consistently outperformed lower learning rates, likely because fine-tuning from COCO weights allows higher initial LR without instability; (2) batch=16 outperformed batch=8 with more stable gradient estimates; (3) conf=0.5 balanced precision and recall — conf=0.3 increased recall to 0.71 but dropped precision to 0.38 with many false positives.

5.4 End-to-End System Performance

Stage	Latency	Notes
YOLO11n detection	<5ms	Per frame on MPS
DVR buffer save	~200ms	Saves last 150 frames as MP4
Gemini video upload	~2s	Depends on network speed
Gemini analysis	~3s	Frame-by-frame analysis of 5-second clip
Browser-use verification	~5s	Autonomous rulebook web search
Total end-to-end	<10s	vs. 2-5 min for traditional VAR

6. Conclusion

We presented Ref Zero, an AI sports referee assistant that combines fine-tuned YOLO11 object detection, MediaPipe pose estimation, Gemini 2.0 Flash video analysis, and autonomous browser-based rulebook verification to deliver VAR-quality decisions in under 10 seconds.

Our fine-tuned YOLO11n model achieves mAP@0.5 = 0.554 on sports foul detection, trained in 2.713 hours on consumer hardware. The ablation study confirms YOLO11n as the optimal choice for real-time edge deployment. The multi-modal pipeline demonstrates that combining specialized detection models with large language models enables nuanced sports officiating beyond simple foul/no-foul binary classification.

Future work includes: expanding the training dataset beyond 353 images via additional Roboflow datasets and synthetic data generation; supporting additional sports (NFL, NHL); implementing multi-camera angle fusion; and deploying drift detection for automated model maintenance as real-world inference distributions shift from training data.

Ref Zero demonstrates that AI can meaningfully assist human referees — not replace them — by providing fast, consistent, and explainable decisions backed by official rulebook citations.

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